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A study on the factors affecting household solar adoption in Kerala, India

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Abstract

Purpose – Recent research on the energy system highlights the need for understanding the bandwidth of drivers and inhibitors of household investor's behaviour in rooftop PV (or photovoltaic power system) and to fit the broader socio-economic context in which they are deployed. However, apart from few exceptions, these newer perspectives have not been duly applied in the research on rooftop PV. This paper aims to fill this gap and to shed new light on rooftop PV investment decisions.

Design/methodology/approach – This study has been conducted with the primary data collected using two data sets of 237 households and 387 households of Indian southern state Kerala using survey-based questionnaire. The findings from first data set revealed that households considering the adoption of PV were likely influenced by six distinct factors, three motivators and three inhibitors. Second data set for multi-state analytic approach was proposed whereby the research model was tested using structural equation modelling (SEM). The outcomes of SEM were used as inputs for an artificial neural network (ANN) model for forecasting investor investment decision in in renewables. The ANN model was also used to rank the relative influence of significant predictors obtained from SEM.

Findings – In line with the risk–return framework, government subsidies act as primary motivator which helps in overcoming the initial risk of investment in the new technology. Further, low prices and low cost of maintenance are some of the financial motivators which may likely mitigate the long-term apprehension of returns and maintenance cost. Lastly, the strongest motivators of PV investment come from the environmental and financial motivator in the form of PV subsidies, which further solidifies the role of policy interventions in investment decision. The ANN model identified the technical barrier and knowledge and awareness factors play a significant role in forcasting the investor investing decision.

Practical implications – The study results will be useful for policymakers for framing strategies to attract and influence their investment in renewable energy.

Originality/value – Building upon behavioural finance and institutional theory, this paper posits that, in addition to a rational evaluation of the economics of the investment opportunities, various non-financial factors affect the household's decision to invest in renewables.

Keywords Private household investment in rooftop PV, Behavioural finance, Risk–return framework in energy investment, Financial motivators and barriers, Environmental, Energy and awareness and institutional factors

Paper type Research paper

Introduction

Sun-powered vitality is a sustainable free wellspring of vitality that is supportable and limitless, dissimilar to limited and non-renewable energy sources. It is likewise a non-contaminating wellspring of vitality, and it does not discharge any ozone-harming substances while creating power. Despite the established multi-fold benefits of solar energy and its widespread diffusion around the world, the private household investment in the rooftop solar remains largely elusive; that too when extensive credit lines, incentive drives, concessional offerings and lucrative loans were made available by governments across the world

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(Masini and Menichetti, 2013). The maximum investment happened in the area of utility solar panels, which are mainly public investments (Stern, 2008). While some scholars agree that rooftop PV (rooftop photovoltaic power system) adoption requires dedicated policies, findings from some other studies point to the contrary (Alvarez *et al.*, 2009; Becker and Shadbegian, 2009). There have been mixed findings in the literature on the efficacy of these dedicated mechanisms. Partially, the failure of these mechanisms and mixed findings in the literature have been attributed to the inability of policymakers and researchers alike in understanding the bandwidth of drivers and inhibitors of the investment decision process and to fit the broader socio-economic context in which they are deployed (Sener *et al.*, 2018; Gimeno *et al.*, 2018; Masini and Menichetti, 2013). Hence, the pointers – limited effectiveness of policy-level mechanisms along with the frivolous understanding of the multitudes of considerations taken by the private investor – suggest the need for a deeper understanding of the processes through which these economic agents arrive at an investment decision on rooftop PV.

The present state continues even after the repeated calls for research. The literature lacks focus on the private household investor of rooftop PV and hence has an underdeveloped understanding of the decision processes (De Leon Barido *et al.*, 2020; Masini and Menichetti, 2012). Further, the focus has been to study the phenomenon at a macro level of the energy market, at the neglect of individual household investor's decisions. Even the studies consider the assumption of rationality while studying the behaviours of investors. This is so when increasing evidence points to the limitations of assumptions of rationality in investment decisions in general and in energy systems specifically. Recent researches in the energy system highlight the need for inclusion of broader context of social and psychological considerations in energy-related investment decisions (Sener *et al.*, 2018; Gimeno *et al.*, 2018; Masini and Menichetti, 2013). Behavioural finance and the bounded rationality landscape have long challenged the validity of the rational-actor models of classical economics in many decision-making contexts. It's only recently that energy economists started giving attention to these newer landscapes, mostly for policy evaluation purposes. However, apart from few exceptions, these newer perspectives have not been duly applied in the research on the household investor's behaviour in rooftop PV and to study the reason behind their lackluster attitude towards PV adoption.

This paper aims to fill in this gap and to shed new light on rooftop PV investment decisions. Building upon behavioural finance and institutional theory, we posit that, in addition to a rational evaluation of the economics of the investment opportunities, various nonfinancial factors affect the household's decision to invest in renewables. We analyse the investment decisions of a large sample of household investors, to identify the main determinants of their choices.

The paper also proposed a new research model for predicting the household investor's behaviour in rooftop PV and examining the key factors that influence the assessment to invest or not to invest in rooftop PV. To examine the proposed research model and questions, two analytic methods (structural equation modelling (SEM) and artificial neural network (ANN)) were applied as suggested by Scott and Walczak (2009). SEM was used for analysing the relationship and strength of independent and dependent factors, whereas the ANN model was used for forecasting investor investment decisions on renewables. ANN model was also used for identifying the relative importance of significant factors, hence, overcoming conventional statistical techniques used for the prediction of consumers' behaviour which usually examine only linear relations among variables.

The present study refers to the psychosocial motivators and inhibitors of the rooftop PV investment decision process to fit the broader socio-economic context in which they are deployed. Investment patterns in solar energy are of special interest, and the present study was done to analyse the behavioural pattern of household investors in the renewable energy investment decision-making process.

Further, the research question bears specific significance in developing countries such as India. Solar energy is an increasingly important part of India's current energy mix and will play a vital role in our clean energy future. As per the Ministry of Commerce and Industry, Government of India, the Indian renewable energy sector is the fourth most attractive renewable energy market in the world. As of October 2018, India was positioned fifth in installing a sustainable power source limit. As per a recent report by PwC and Climate investment funds (2019, p. 5), "despite all the underlying benefits, rooftop solar in India has not achieved significant growth. Various international credit lines and concessional funding have supported the market growth in recent years by reducing the high cost of financing for smaller projects, thereby reducing the tariffs and making the projects more viable. However, there still exist many challenges in this segment, such as awareness building, lack of capacity, legal and contractual issues, and roof right issues, which need to be addressed at each stakeholder level to help further scale up the deployment."

Theoretical underpinning

Decades of research on behavioural finance established that investors take decisions with limited information, by acting under bounded rationality. Then the question is what are the implications of behavioural finance and bounded rationality literature on renewable energy investment and policy?

One of the important theoretical implications of bounded rationality perspective is the influence of path dependence on new investment decisions, means past decisions and events could impact on present choices. For example, Wustenhagen and Menichetti (2012, p. 5) have identified evidence for path dependence in the "venture capital market, which slowed down the flow of venture capital investments into the newly emerging renewable energy sector; discuss effects of path dependence on Finland's attempts to diversify its economy away from fossil fuels; it can also lead to lock-in, and it has been suggested that the world is currently locked-into a high-carbon economy, which is difficult to transcend" (Lovio *et al.*, 2011). Path dependence precludes oil companies to shift from the previous path of carbon energy to renewable energy which is less trodden.

The past literature on path dependence suggests that the answer lies in the statement "rather than some objective measure of risk and return, a behavioural finance perspective would suggest that perceptions matter, and that perception of risk and return" in new investment "are influenced by cognitive factors" which play a critical role in breaking the path from previous investment (Wustenhagen and Menichetti, 2012, p. 4). These psychosocial behavioural factors bring individual investors into the centre, where non-financial and behavioural factors play an important role in breaking path dependence.

Theorists and behavioural economists have questioned the rational-actor models of classical economics and proposed that "cognitive and cultural factors, as well as personal beliefs, affect an individual decision in the renewable energy literature" and have started "considering the role of personal beliefs especially to study path-dependent barriers" to the diffusion of rooftop PV, and "broader range of social and personal factors affecting human interactions with social and policy institutions may act as enablers" in overcoming path dependence to rooftop PV (Masini and Menichetti, 2013, p. 7). Therefore, the present study refers to the path-dependent psychosocial inhibitors and motivators of overcoming path dependence in traditional carbon-based investment to the rooftop PV investment decision process and to fit the broader socio-economic context in which they are deployed.

Review of literature and hypothesis development

In Santiago, Chile, the economic situations are magnificent for the family unit selection of PV innovation, yet the take-up is unimportant. To investigate this conundrum, the creators led a Delphi study concentrated to request the information of a board of Chilean PV specialists. Further, the research question bears specific significance in developing countries such as India. Solar energy is an increasingly important part of India's current energy mix and will play a vital role in our clean energy future. As per the Ministry of Commerce and Industry, Government of India, the Indian renewable energy sector is the fourth most attractive renewable energy market in the world. As of October 2018, India was positioned fifth in installing a sustainable power source limit.

Previous literature on domestic PV diffusion has shown that "barriers and enablers tend to go beyond technical and policy advancement" and instead reside at the intersection of "social, economic, and technical context" (Sovacool, 2014; Popp *et al.*, 2011). In Senegal, for example, Thiam (2011) demonstrated the importance of "social acceptance of renewable technologies". As per Walter *et al.* (2018, p. 126) "adoption has been thwarted by consumer perceptions of solar-panel aesthetics; residents considering adopting solar panels have also been influenced by other peer-related effects, such as convincing conversations between neighbors and solar PV owners, resulting in solar panel owners being clustered together" (Rai *et al.*, 2016; Graziano and Gillingham, 2015; Richter, 2014; Rai and Robinson, 2013; Faiers and Neame, 2006; Margolis and Zuboy, 2006).

Another key role in diffusion may also be played by consumer knowledge "customers were strongly influenced, for example, by their ignorance of PV technology, including issues related to permitting, planning, and maintaining it over its lifecycle" (Walter *et al.*, 2018, p. 128; Beck and Martinot, 2012; Margolis and Zuboy, 2006). Some studies have identified socio-technical barriers that "some consumers disregard renewable energy sources because they do not know of its economic benefits or their energy consumption, or because their paradigms of permissible energy sources point them to traditional fossil-fuel power plants" (Walter *et al.* 2018, p. 130; Sovacool, 2009; Margolis and Zuboy, 2006).

Whether consumers adopt solar panels is also influenced by socio-economic barriers, they perceive some "financial risk and are hesitant to invest in solar PV", a hesitance reinforced by the "relatively high capital costs of installation may even be exacerbated by competing for energy providers who offer subsidies for fossil fuels" (Walter *et al.*, 2018, p. 129; Rai *et al.*, 2016; Beck and Martinot, 2012; Rai and McAndrews, 2012; Faiers and Neame, 2006; Popp *et al.*, 2011). Various other researches point to other socio-economic considerations including "high capital and maintenance costs; limited experience with new energy technology; as well as of under-valuing the long-term benefits of environmental investments" (Masini and Menichetti, 2013, p. 7).

Consumers must also contend with their lack of trust in available information on PV, "uncertainty of the costs associated with operation and maintenance" and "the long timeline for return on investment (ROI)" (Rai *et al.*, 2016, p. 502; Rai and Robinson, 2013; Rai and McAndrews, 2012). Some other studies have shown that due to path dependence of previous carbon-based technology, "consumers must see that the present value of a PV investment greatly exceeds the investment cost by a factor that can surmount consumer uncertainty" (Bauner and Crago, 2015, p. 31). All these fluctuations lead to individual investors going away from the adoption of PV "until market and policy maturity reach a satisfactory level of perceived risk, known as the optional value" (Bauner and Crago, 2015, p. 31).

Additionally, on account of Bangladesh, one examination showed that different variables decide a client's fulfilment for the utilization of a sun-oriented home framework (Komatsu *et al.*, 2013). In the investigation, the client's impression of the advantages of the sunlight-based home framework brought about a larger amount of adopter's fulfilment saw benefits incorporate the nature of gear and change in the family unit way of life, for example, an expansion in youngsters' examination time. A few examinations have explored the impact of friend consequences for family unit sunlight-based vitality innovation reception choice (Rai *et al.*, 2016). Another examination found a beneficial outcome of the state approach impetus on PV sunlight-based innovation selection (Crago and Chernyakhovskiy, 2017). In a similar manner, an examination on Pakistani families discovered proof that the money-related help by the administration for the establishment of little sun-powered PV frameworks influenced the appropriation choices (Qureshi *et al.*, 2017).

These endeavours yielded 26 factors – the two inspirations and hindrances – affecting the dispersion of PV. Of the 26, specialists agreed on the overall significance of 21. The writing recommends that dispersion of PV advancements is affected by complex specialized, financial and social components. Essentially, the specialists saw the impact of money-related, ecological and vitality supply factors. They saw rising hindrances to appropriation as being monetary, specialized and institutional and information factors. They considered the most essential elements impacting selection to be monetary inspirations and budgetary boundaries. They viewed the least imperative factors as natural inspirations and specialized hindrances. With this information, the creators build up a selection system for family PV that portrays the communication among the recognized inspirations and boundaries. This structure educates strategy suggestions for Santiago, Chile, and adds to the collection of writing investigating the interconnected frameworks of variables that impact common foundation all in all and PV selection (factors influencing household solar adoption in Santiago, Chile, by (Walters *et al.*, 2018)).

An increasingly relevant body of literature is investigating the role of public regulators, by looking at how policies should be designed to stimulate the growth of renewable energy innovations. Researchers have broadly questioned the viability of various arrangements of emotionally supportive networks in accomplishing approach goals. For instance, (Menanteau *et al.*, 2003) have featured the positive job of feed-in levies in prompting an expansion of the sustainable power source share by bringing down the hazard related to the speculation of choice. Other studies provide support to the hypothesis that feed-in tariffs are a more effective policy scheme, compared to market-based approaches (Butler and Neuhoff, 2008). In contrast, other analyses indicate that the hidden costs of feed-in tariffs may offset their benefits by determining liabilities in the long term (Liebreich, 2009).

Different examinations accentuated the natural, financial, social and political determinants of family unit private PV selection conduct (Davidson *et al.*, 2014; De Groote *et al.*, 2016). In China, motivations, for example, the feed-in levy were recognized as the basic driver of household PV sun-oriented innovation organization (Zhang *et al.*, 2007). Another examination on selection and non-selection of sun-powered PV in the Netherlands discovered four basic drivers, in particular, "the apparent relative preferred standpoint of the innovation, the multifaceted nature of the advancement, social influence, and knowledge of grants and costs" (Vasseur *et al.*, 2010, p. 891).

Crago and Chernyakhovskiy (2017) analysed the adequacy of state arrangement motivators connected in the United States to increment private sun-powered PV limit. The investigation found that sun-powered PV request was emphatically influenced by money-related motivating forces that decrease the direct front expense of sun-oriented PV appropriation, expert ecological inclination and sunlight-based explicit commands.

Determinants of household adoption of solar energy technology

Renewable energy investment and financial performance: Considering the social parts of the individual basic leadership process is imperative in distinguishing and tending to hindrances in the selection of residential solar PV. In any case, there is minimal efficient research concentrating on these parts of private PV in Texas, an imperative, substantial, crowded state, with a scope of difficulties in the power division including expanding request, shrinking hold edges, obliged water supply and testing discharges decrease focussing under proposed government guidelines. This paper means to address the gap through an observational examination of another study put together data set gathered in Texas to sun-powered vitality in a moderately undiscovered market and help recognize data gaps that could be focussed to lighten key boundaries to embracing sun-oriented, along these lines empowering critical discharges hence it decreases in the emission in Texas (Rai and Beck, 2015).

Studies on the comparison of the performance of renewable energy investments versus traditional assets are also becoming common. These studies showed that, in recent years, capital investment in renewable energy technologies has led to superior performance compared to more traditional investments (Deutsche bank, 2009). Explaining this scenario, by clarifying the underlying relations between solar energy investments and capital performance would represent a useful theoretical contribution and it would also help the government design more effective policies to attract further investments in this area. Unfortunately, management scholars have somewhat overlooked this topic so far. Given the absence of concentrates explicitly tending to the connection between sustainable power source ventures and monetary execution, some helpful bits of knowledge can be acquired from concentrates in related fields, for example, those looking at the connection between natural execution and money-related execution. (Hart and Ahuja, 1996).

An increasingly relevant body of literature is investigating the role of public regulators, by looking at how policies should be designed to stimulate the growth of renewable energy innovations. Scholars have extensively disputed the effectiveness of different policy support systems in achieving policy objectives. For instance, (Finon and Menanteau, 2003) have highlighted the positive role of feed-in tariffs in leading to an increase in the renewable energy share by lowering the risk associated with the investment decision. Other studies provide support to the hypothesis that feed-in tariffs are a more effective policy scheme, compared to market-based approaches (Neuhoff, 2005). In contrast, other analyses indicate that the hidden costs of feed-in tariffs may offset their benefits by determining liabilities in the long term (Liebreich, 2009; Bazilian *et al.*, 2013).

Investments in renewable energy technologies have increased steadily over the past years. Experts estimated from 2002 until the finish of 2009 that the environmentally friendly power vitality advertisement has pulled in more than US$650*bn* in total (Masini and Menichetti, 2012). The years 2006 and 2007 have encountered twofold digit development, while 2008 and 2009 have been influenced by the outcomes of the worldwide monetary emergency, which has affected the spotless vitality advertising both specifically (through a liquidity crush) and in a roundabout way (using a general fall in worldwide vitality request related with lower oil and gas costs). Even though the financial downturn has hit the sustainable power source showcase extremely, specialists trust that in the long haul the perfect vitality segment ought to recuperate preferably and quickly over others. Numerous administrations have propelled green financial estimates that have added to direct the fall of investments in this industry. Besides, the IEA as of late required a “Clean Energy New Deal” to abuse the money-related and monetary emergency as a chance to prompt a lasting movement in contributing to low-carbon advancements (IEA, 2009) (Olivier and Peters, 2009).

Arrangement targets and measures have been set by a few nations worldwide to help the organization of sustainable power source innovations. Over 80 nations have effectively settled sustainable power source approaches. Strategies can assume a vital job in diminishing the hazard related to a speculation choice, by giving a steady system and diminishing uncertainty in the market. As pointed out by Ecofys (2008) (De Jager *et al.*, 2008) “commitment, stability, reliability, and predictability are all components that expansion certainty of market performers, diminish administrative dangers, and thus fundamentally decrease the expense of capital".

As stated by Shleifer (2000, p. 181) (A, 2000), "the perception of risk is one of the most intriguing open areas in behavioural finance". The author highlights the need to develop a conceptual model to fully capture the way investors assess risk, their rules of thumb and how they forecast expected scenarios. In turn, (Thaler, 1996) points out that adding a human element to financial market analysis can lead to a better understanding of the mechanisms underlying market behaviour.

An increasingly relevant body of literature is investigating the role of public regulators, by looking at how policies should be designed to stimulate the growth of renewable energy innovations. Researchers have broadly questioned the viability of various arrangements of emotionally supportive networks in accomplishing approach goals. For instance, (Menanteau *et al.*, 2003) have featured the positive job of feed-in levies in prompting an expansion of the sustainable power source share by bringing down the hazard related to the speculation of choice. Other studies provide support to the hypothesis that feed-in tariffs are a more effective policy scheme, compared to market-based approaches (Butler and Neuhoff, 2008). In contrast, other analyses indicate that the hidden costs of feed-in tariffs may offset their benefits by determining liabilities in the long term (Liebreich, 2009).

Although solar power systems are considered as one of the most promising renewable energy sources, some uncertain factors, as well as the high cost and uncertainty on ROI, could be barriers that create customer resistance. Leasing instead of purchase, as one type of product-service system, could be an option to reduce consumer concern on such issues. Households think that there is a price variation for the same solar panels from different suppliers. There is a difficulty for households to compare costs between different solar panels.

H1. Financial motivation is positively associated with higher investment in rooftop PV.

H2. Financial barriers are negatively associated with investment in rooftop PV.

Renewable energy investment and energy–environment motivation: Based on the study on the factors determining household use of clean and renewable energy sources for lighting in Sub-Saharan Africa, it reveals that rural households are dependent largely on kerosene and batteries for lighting their houses, while electricity and batteries form a major source of energy for lighting urban households

In light of the examination, putting resources into environmental change: an advantage management viewpoint the creator contends that the developing venture opportunity in environmental change was driven by long-term megatrends that would proceed into the not so distant. The examination in this report is driven by two objectives of need and opportunity. They have another test: how to discover the financing to create and send advancements they have to moderate and adjust to environmental change.

Environmental problems and health risks stemming from the household use of traditional fuels remain a significant challenge for households living in poverty in developing countries. This paper applies a logit model to dissect the driving components of a family's choice to the selection of sun-based vitality innovation. To do this, the examination utilizes cross-sectional information gathered from two kebeles in the areas of Woliso, a town of Oromia local state in focal Ethiopia. The outcomes uncover that an incredible number of variables were found to have a beneficial outcome on sun-powered vitality innovation selection. The discoveries of this paper build up that well-off and increasingly instructed family units are bound to embrace sun-powered vitality innovation contrasted with more unfortunate ones. Male-headed family units are less inclined to receive sunlight-based vitality innovation contrasted with female-headed partners (Guta, 2018).

Crago and Chernyakhovskiy (2017) analysed the adequacy of state arrangement motivators connected in the United States to increment private sun-powered PV limit. The investigation found that sun-powered PV request was emphatically influenced by money-related motivating forces that decrease the direct front expense of sun-oriented PV appropriation, expert ecological inclination and sunlight-based explicit commands. Although the setting in creating nations is unique, monetary motivating forces for diminishing the direct front expense of sun-oriented PV reception are dared to assume a fundamental job of forming family unit conduct. This is because the expense of sun-based PV remains a critical variable that stops the more extensive dispersion of sun-powered PV framework (Qureshi *et al.*, 2017). As per the experts' opinion, there are some motivational factors such as financial reasons, environmental reasons and energy supply reasons. Renewable energy can contribute to reducing the global warming. But for average investors, they often fall in emotions rather than using their logic. That is why there is relevance in understanding the behavioural factors affecting the investment decision-making process in renewable energy as it can contribute to the sustained energy system.

The easiest and best approach to increase some vitality freedom is to introduce sun-powered PV. It can produce your capacity and fare the overabundance to the framework, for which you will get a credit. Aside from straightforward upkeep like clockwork, there is nothing to check or stress over – essentially appreciate the diminished power bills. Introducing a solar panel is a money-related venture and more often than not mortgage holder's fundamental concern will be its toughness just as proficiency. One advantage of sun-oriented board establishment is that the frameworks are entirely dependable. They have ended up being trustworthy wellsprings of vitality. Solar energy creates clean, renewable power from the sun and benefits the environment. Alternatives to fossil fuels reduce carbon footprint at home and abroad, reducing greenhouse gases around the globe. Solar is known to have a favourable impact on the environment.

H3. Energy motivation is positively associated with higher investment in rooftop PV.

H4. Environmental motivation is positively associated with higher investment in rooftop PV.

Renewable energy investment and awareness–technical barriers: (Schelly, 2014) contemplated the elements that influence the family choice for sun-oriented vitality innovation appropriation. The creator underlined that family unit appropriation choice is formed by the biophysical and monetary structure as well as by an intricate arrangement of direction for living, the transaction of pointers of family unit conditions and access to data. Additionally, on account of Bangladesh, one examination showed that different variables decide a client's fulfilment to the utilization of a sun-oriented home framework (Komatsu *et al.*, 2013). In the investigation, the client's impression of the advantages of the sunlight-based home framework brought about a larger amount of adopter's fulfilment saw benefits incorporate the nature of gear and change in the family unit way of life, for example, an expansion in youngsters' examination time. A few examinations have explored the impact of friend consequences for family unit sunlight-based vitality innovation reception choice (Rai *et al.*, 2016). Another examination found a beneficial outcome of the state approach impetus on PV sunlight-based innovation selection (Crago and Chernyakhovskiy, 2017). In a similar manner, an examination on Pakistani families discovered proof that the money-related help by the administration for the establishment of little sun-powered PV frameworks influenced the appropriation choices (Qureshi *et al.*, 2017).

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There would appear, however, to be some time lag between the initiation of emissions reduction efforts and the realization of "bottom line" benefits. Firstly, pollution prevention requires an upfront investment in training and equipment. Just as in TQM, pollution prevention appears to require the proper bounding of the system or process to be improved, the involvement of key stakeholders in the process and the establishment of clear goals and targets for improvement (Ishikawa and Lu, 1985; Imai, 1986). Secondly, the savings from emissions reduction may take some time to be realized as the renegotiation of supply and waste disposal contracts and internal reorganization may be required (White *et al.*, 1993). For example, cutting emissions by 50% may mean that dramatically less raw material is needed. Supply contract renegotiations may be required to reduce the inventory to realize the cost savings. Similarly, drastically lower emissions and waste may also mean renegotiating waste disposal contracts or storage and treatment arrangements. Finally, to realize savings in compliance costs, it may be necessary to reassign or reduce internal personnel involved in legal compliance or pollution control activities (Hunt and Auster, 1990).

Similarly, as the quantity of utility sun-powered activities develop, there is likewise developing enthusiasm for utilizing housetop sun-powered boards and another little scale, nearby power sources known as dispersed age (DG). To empower the presentation of these frameworks when they originally came to showcase years prior, numerous states affirmed a charging framework called net metering. While net metering arrangements fluctuate by state, users with housetop sun-oriented or other DG frameworks, as a rule, are credited at the full retail electric rate for any overabundance power they create and pitch to their neighbourhood electric utility using the electric lattice. Electric utilities are required to purchase this power, even though it, for the most part, would cost them less to deliver the power themselves or to purchase the power on the discount showcase from other power suppliers.

Net metering is a charging framework that enables electric users to pitch to their electric utility any overabundance power produced by their DG frameworks. A wide range of DG sources might be qualified for net metering credits; however, housetop sun-based establishments are the most well-known kind of DG advanced with net metering. While net metering approaches change by state, users with housetop sun-based or other DG frameworks, as a rule, are credited at the full retail power rate for any power they pitch to electric utilities through the lattice. The full retail power rate incorporates the expense of the power as well as the majority of the fixed expenses of the posts, wires, metres, cutting-edge innovations and other foundation that makes the electric framework sheltered, dependable and ready to suit sunlight-based boards or other DG frameworks. Through the credit or installment they get, net-metered users adequately abstain from paying these expenses for the framework.

Solar systems in Kerala delegated into two separate classifications: off-grid solar system and on-grid solar systems. Off-grid solar power systems enable you to store your sunlight-based power in batteries for use when the power matrix goes down or on/off chance that you are not on the framework. The advantage of off-framework heavenly bodies is that they can give the capacity to your loads when the power grid is down.

On-grid solar power systems can create control when the utility power matrix is accessible. They should interface with the grid to work. They can send overabundance control produced back to the matrix when you are overproducing so you credit it on the bill. The primary impediment is you need to confront the successive power cuts in Kerala. Since once the matrix is disconnected, there are no other sources for your capacity prerequisites when you are utilizing an on-network close planetary system. Solar panels do not have enough offers or promotions so that every household can consume it.

Solar panels do not require much maintenance. They are entirely tough and should last around 25–30 years with no upkeep. The main support need to perform is to wash them clean and residue two to four times each year, which can without much of a stretch do with a patio nursery hose. This essential cleaning routine guarantees that the sun can sparkle brilliantly on the board, augmenting the measure of light accessible to transform into electrical power. Yet, family units are missing appropriate information on the support.

Notwithstanding cost, while picking the best sun-oriented board for establishment circumstance, it is vital to consider both how it is fabricated and what materials are utilized. There are three levels of product quality and energy matters just supply sun-powered boards from the primary level. Level 1 incorporates the best 2% of sunlight-based PV (photovoltaic) board makers. Level 1 producers are those that are vertically incorporated. This implies they control each phase of the assembling procedure. These organizations put vigorously in innovative work use of progressed mechanical procedures and have been producing sun-powered boards for over five years.

Level 1 makers utilize the most perfect and best grade of silicon to deliver sunlight-based cells. The higher the silicon grade, the more extended the sun-based cell will last, and the better it will perform in changing over the sun's vitality to power.

Level 2 incorporates organizations that put less in innovative work, that are dependent on both mechanical and manual get together on generation lines and have frequently been in sun-oriented board make for 2–5 years. These tier 2 makers can create great boards at great costs, yet it very well may be a hit and miss issue.

Level 3 includes 90% of new sun-based PV producers. These organizations collect boards just, they typically do not make their phones and do not put resources into innovative work. While frequently accessible at a less expensive value, level 3 makers utilize human generation lines for manual patching of sunlight-based cells, which regularly isn't the best methodology as quality can differ from administrator to administrator and every day. A tier 3 sun-oriented board may over the long haul cost you substantially more as far as unwavering quality and power yield.

Households do not think that they are getting access to quality materials. Each proficient organization must offer savvy sun-powered arrangements that spare a great deal of cash. Family units believe that the majority of the solar companies give the close planetary system, which does not ensure the future prerequisites. When anticipating the overhaul later on you may finish up changing the equipment and in it prompts misfortune.

When it comes to residential solar installations, solar panels are often found on sloped rooftops. There are many mounting framework alternatives for these calculated rooftops, with the most widely recognized being railed, rail-less and shared rail. These frameworks require some kind of infiltration or securing into the rooftop, regardless of whether that is connecting to rafters or specifically to the decking. Households feel that roof mounting cannot damage the user's roof and cause leaks into the house. There is an aesthetic belief that solar panels can cause an ugly look at their house in which the households are not agreeing.

H5. Knowledge and institutional barriers are negatively associated with investment in rooftop PV.

H6. Technical barriers are negatively associated with investment in rooftop PV.

Methodology used

Using a multiple-analytic approach as proposed by Scott and Walczak (2009), both SEM and ANN modelling were employed in this research. SEM inspects the reliability and validity of the measures and tests the proposed hypotheses. ANN was used for identifying the importance of the variables as well as further certified the antecedents as predictors of investment in rooftop PV decision. ANN is considered as a good tool for prediction, but having limitation is its "black box" approach, making it unsuitable for hypotheses testing and examining causal relationships. This can be overcome by using SEM approach. Aiming to understand the investment in the rooftop PV decision of the people, this study was divided into two parts.

Study 1 is used for categorizing antecedent factors influencing investment decisions amongst the Indian investors. The objective of study 1 is to verify the factors in the Indian context using exploratory factor analysis. The finding of study 1 is used for Study 2 for testing the proposed hypothesis and identifying the importance of the variables.

Measurement scale

For measuring the key constructs for this research, the scales were adopted from the existing literature. To examine investors' motivation for financial motivation (FM) for the adoption of solar panels, Walters *et al.*'s (2018) scale was amended. The sample item includes "Lack of household knowledge on proper maintenance". Other variables of investors' motivation were sketched from the various studies – energy supply motivation Walters *et al.* (2018); Masini and Menichetti (2013); Wüstenhagen and Menichetti (2012), sample item includes "I am highly motivated in the adoption of solar panels due to energy independence" and environmental motivation (Walters *et al.*, 2018), sample item includes "Family desire to protect the environment through the use of solar panels". For measuring the perception of barriers such as financial barriers, (Rai and Beck, 2015; Walters *et al.*, 2018) study was used, sample item includes "Solar panels have high installation cost". Other barriers variables are institutional and technical barriers scales adopted from (Rai and Beck, 2015; Walters *et al.*, 2018), sample item includes "Roof mounting can damage the clients' roof and cause leaks into the house".

Consumers' knowledge and awareness were measured using Schelly (2014); Qureshi *et al.* (2017) scale consisting of a sample item "Scarcity of attractive offers, promotion provided to the potential buyers". Lastly, to observe the investor investment decision, the scale developed by Rai and Beck (2015); Walters *et al.* (2018) was used. The pool of experts from academia and industry were hired for scrutiny of these questionnaires. After trivial reformations in the language of statements, the final survey instrument comprised of 25 items covering aforestated constructs. A five-point Likert scale was used to measure the extent to which respondents agreed or disagreed. The anchors here included 1 and 5 denoting "Strongly disagree" and "Strongly agree", respectively.

Study 1 – categorize antecedent factors influencing investment decisions amongst the Indian investors

As mentioned earlier a questionnaire comprising of 25 items was developed, and this questionnaire was shared with 237 normal and institutional households of Kerala. All normal households must be Pucca house (Pucca housing refers to dwellings that are designed to be solid and permanent) and institutional households as per the Census of India (2010) definition were selected for data collection. These houses were selected because a resident of these house owners was considered to be above the poverty line and can invest the money on rooftop PV. The list of 25 elements was split into two sections – the first section comprises of motivation questions and the second had the remaining items related to barriers and knowledge. After removing the incomplete or unfilled survey forms, a total 201 data sets were used for analysis. To assess the preliminary factor structure related to investor investment decision, exploratory factor analyses were conducted. Principal component analyses were used for segregate and reducing the number of items by removing the items with communality value less than 0.50 and/ or that had cross-loading.

Removing nine items from the questionnaire yielded six factors solution – knowledge and awareness, financial, environment and energy motivation, financial barriers and technical barriers, accounting for 71% of total variance explained (Table 1).

The idea of rotation is to reduce the number of factors on which the variables under investigation have high loadings. Looking at the table we can see that difficult to understand the net billing framework, hard to find information, scarcity of attractive offers, proper maintenance, lack of access to quality material and lack of knowledgeable companies are substantially loaded on factor 1 which is knowledge and awareness while energy reliability, energy independence, initial subsidies, protection of the environment are substantially loaded on factors 2, 3 and 4, respectively, which is motivation while uncertainty on ROI, price variation, comparison of costs are loaded on factor 3 which is financial barriers. Roof mounting and aesthetic beliefs are loaded on factor 4, which are technical barriers.

Factor 1: Knowledge and institutional barrier: As the quantity of utility sun-powered activities develop, there is likewise developing enthusiasm for utilizing housetop sun-powered boards and another little scale, nearby power sources. Solar panels do not require much maintenance. They are entirely tough and should last around 25–30 years with no upkeep. Yet, family units are missing appropriate information on the support. Households do not think that they are getting access to quality materials. Family units believe that the majority of the solar companies give the close planetary system, which does not ensure the future prerequisites.

Factor 2: Energy motivation: The easiest and best approach to increase some vitality freedom is to introduce sun-powered PV. Introducing a solar panel is a money-related venture and more often than not a mortgage holder's fundamental concern will be its toughness just as proficiency. One advantage of sun-oriented board establishment is that the frameworks are entirely dependable. They have ended up being trustworthy wellsprings of energy vitality.

Factor 3: Financial motivation: The idea of capital sponsorships is extremely basic – the administration gives back a certain % of the forthright capital expense. Along these lines, if your complete capital expense is Rs 1 lac and the administration gives back 30%, your compelling capital expense is just Rs 70,000 (1 lac – 30%* 1 lac). Family units are particularly inspired to embrace sun-powered boards because of starting sponsorships.

Factor 4: Environment motivation: Solar energy creates clean, renewable power from the sun and benefits the environment. Alternatives to fossil fuels reduce carbon footprint at home and abroad, reducing greenhouse gases around the globe. Solar is known to have a favourable impact on the environment.

Factor 5: Financial barriers: Although solar power systems are considered as one of the most promising renewable energy sources, some uncertain factors, as well as the high cost and uncertainty on ROI, could be barriers that create customer resistance. Leasing instead of purchase, as one type of product-service system, could be an option to reduce consumer concern on such issues. Households think that there is a price variation for the same solar panels from different suppliers. There is a difficulty for households to compare costs between different solar panels.

Table 1. Rotated component matrix

	Component					
	Knowledge and awareness	Energy motivation	Financial motivation	Environmental motivation	Financial barriers	Technical barriers
Existing net billing framework is difficult to understand and engage with	0.784					
Hard to find information on how to interpret the benefits of excess energy sold back to the grid based on law	0.716					
Scarcity of attractive offers, promotions provided to potential buyers	0.692					
Lack of household knowledge on proper maintenance	0.656					
Lack of knowledgeable companies for consulting/distribution/installation	0.641					
Lack of access to quality material or installation practices	0.623					
I am highly motivated in the adoption of solar panels due to energy reliability		0.81				
I am highly motivated in the adoption of solar panels due to energy independence		0.795				
Price of the normal solar grid is affordable enough			0.763			
I am highly motivated in the adoption of solar panels due to the initial subsidies given by the government			0.754			
Family desire to protect the environment through the use of solar panels				0.769		
Family desire to reduce their carbon footprint through the use of clean energy				0.669		
Solar panels have high installation cost					0.818	
There is a price variation in the same solar panels from different suppliers					0.662	
There is an uncertainty regarding return on investment					0.658	
Roof mounting can damage the clients' roof and cause leaks into the house						0.885
Homeowners think that solar PV systems make their house look ugly						0.85

Factor 6: Technical barriers: When it comes to residential solar installations, solar panels are often found on sloped rooftops. There are many mounting framework alternatives for these calculated rooftops, with the most widely recognized being railed, rail-less and shared rail. Households feel that roof mounting cannot damage the user's roof and cause leaks into the house**.** There is an aesthetic belief that solar panels can cause an ugly look at their house in which the households are not agreeing.

Study 2 – examining the predictive validity

To validate the factors and assess its predictive validity, Study 2 was conducted for the proposed model as shown in Figure 1. Non-probability technique – purposive sampling was adopted to minimize the research time and cost (Babbie, 2010). All the nine items that were removed in factor analysis were dropped in the second study during the data collection process. The people above 18 years of age comprised the population. The study has been done with 387 samples. Samples were collected from the households of Kerala. Excluding 56 incomplete questionnaires, a total of 331 responses were used for data analysis.

Besides the data needed for the study, information about the respondents' demographics – gender, age, income and education qualification was also solicited. The final sample comprised of 58.61% males and 41.39% females. Around 56.19% of respondents were in the age group of 20–40 years while the rest were above 40 years. The top three ranges of the respondents' income were – INR15,00,001 and above (27.19%); (b) INR 3,00,001–5,00,000 (24.77%) and (c) INR 1,00,000–3,00,000 (18.73%).

Measurement model testing

The first step in predictive validity using SEM is confirmatory factor analysis (CFA) as used previously by numerous other studies (Mittal, 2019; Mittal *et al.*, 2019). It was used for checking the reliability and validity of the measurement model. The reliability of the model

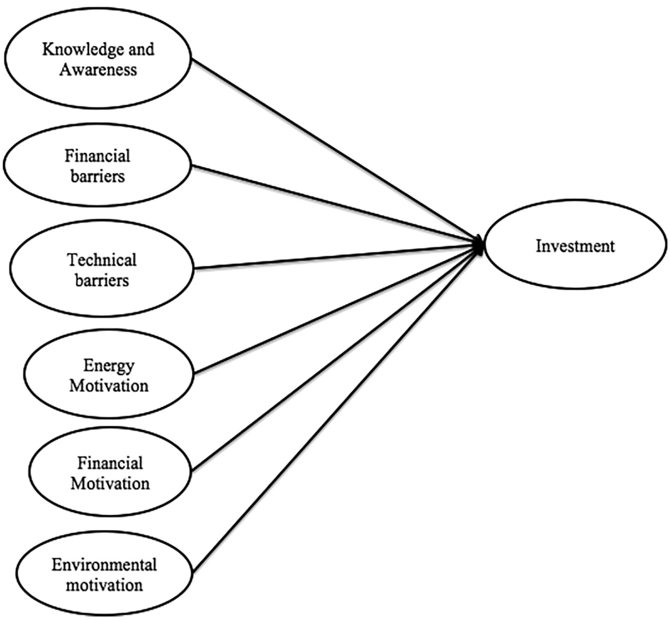

Figure 1.
Proposed model

was checked using composite reliability (CR) criteria, As shown in Table 2, all CR values are higher than 0.70 which reflects the reliability of a model (Parsad *et al.*, 2019). For assessing the convergent validity two criteria were used – first is item loading (more than 0.50); and average variance extracted (AVE) (more than/ equal to 0.50) (Hair *et al.*, 2010). These all suggest a high level of reliability and validity of the measurement model. The square root of AVE for each construct is noted to be greater than the inter-construct correlations (See Table 3). This confirms the discriminant validity (Hair *et al.*, 2010).

The goodness-of-fit statistics confirmed the proposed model to be a good fit for the data. The ratio of Chi-square minimum to the degree of freedom (CMIN/DF) being 1.418 is below the cut-off criterion of 3.00 (Hair *et al.*, 2006). This also indicates a good fit between the hypothesized model and the data. Other indices such as goodness-of-fit index (GFI), comparative fit index (CFI), incremental fit index (IFI) and Tucker–Lewis index (TLI) should be greater than 0.9, and the root mean square error of approximation (RMSEA) should be less than 0.08 (Hair *et al.*, 2006). From the study, fit indices were found to be within acceptable ranges (GFI = 0.953; CFI = 0.979; IFI = 0.977; TLI = 0.971 and RMSEA = 0.036).

Construct	Item	Standardized loading estimates	AVE	CR
Financial motivation (FM)	I am highly motivated in the adoption of solar panels due to the initial subsidies given by the government	0.807	719	0.836
	Price of the normal solar grid is affordable enough	0.887		
Energy motivation (ENM)	I am highly motivated in the adoption of solar panels due to energy reliability	0.768	0.668	0.801
	I am highly motivated in the adoption of solar panels due to energy independence	0.864		
Environmental motivation (EVM)	Family desire to protect the environment through the use of solar panels	0.786	0.699	0.822
	Family desire to reduce their carbon footprint through the use of clean energy	0.883		
Technical barrier (TB)	Roof mounting can damage the clients' roof and cause leaks into the house	0.944	0.694	0.816
	Homeowners think that solar PV systems make their house look ugly	0.705		
Financial barrier(FB)	Solar panels have high installation cost	0.873	0.561	0.784
	There is a price variation in the same solar panels from different suppliers	0.501		
	There is an uncertainty regarding return on investment	0.82		
Knowledge and institutional barrier (KNINB)	Lack of household knowledge on proper maintenance	0.636	0.501	0.855
	Lack of knowledgeable companies for consulting/distribution/installation	0.808		
	Scarcity of attractive offers, promotions provided to potential buyers	0.791		
	Hard to find information on how to interpret the benefits of excess energy sold back to the grid based on law	0.689		
	Existing net billing framework is difficult to understand and engage with	0.677		
	Lack of access to quality material or installation practices	0.613		

Table 2. Factor loading, reliability and convergent validity

	FM	KNINB	FB	TB	ENM	EVM
FM	*0.848*					
KNINB	0.550	*0.706*				
FB	−0.027	0.021	*0.749*			
TB	−0.233	−0.106	0.289	*0.833*		
ENM	−0.009	0.147	0.384	−0.040	*0.817*	
EVM	0.584	0.529	0.023	−0.085	0.056	*0.836*

Note(s): Diagonal values are the square root of AVE; off-diagonal values are the inter-construct correlations. AVE = average variance extracted; FM: Financial Motivation, KNINB: Knowledge and Institutional Barrier, FB: Financial Barrier, TB: Technical Barrier, ENM: Energy Motivation, EVM: Environmental Motivation

Table 3.
Discriminant validity

Structural equation modelling

After the confirmation of the reliability and validity of the elements, the authors undertook a single-stage analysis. Using AMOS, both measurement and structural models were simultaneously estimated. As required in structural modelling, χ^2/DF value is evaluated first. Here, the χ^2/DF value arrived is 1.245, which is significant ($p < 0.001$). Other fit indices – GFI (0.953); AGFI (0.929); CFI (0.986); CMIN/df (1.475); IFI (0.986); TLI (0.981); and RMSEA (0.028) also reflect an acceptable fit for the proposed model.

Hypothesis testing

According to hypothesis H5, it was assumed that knowledge and Institutional barrier (KNINB) of an individual would have a noteworthy influence on investment decisions. From Table 4, it is noted that FB is negatively related to investment decision ($\beta = -0.227$, $p = 0.039$), supporting hypothesis H2. This specifies that the financial barrier would reduce the tendency for investing in green energy. This is in line with the finding of earlier studies such as Walters *et al.* (2018). Similarly, the relationship of technical barrier with investment decision (H6) is also found significant (see Table 4).

As per H3, it was proposed that energy motivation (ENM) would positively influence individuals' investment decisions. The effect of ENM is found significant ($\beta = 0.146$, $p = 0.083$), signifying the importance of ENM aspects for encouraging investment in green energy.

The next two hypotheses H1 and H4 proposed that investment decisions of individuals' on green energy sources are positively influenced by both FM (H1) and environmental motivation (EVM) (H4). The findings also projecting that both FM and EVM have a significant impact on the investment decision on green energy, supporting the first and fourth hypothesis.

Neural network analysis

Derived from mathematical neuroscience, the ANN is a combination of "interconnected predictor variables, where each variable is attached with weightage" (Rumelhart *et al.*, 1986).

Factors	β	Stan. error	p-value	Result
Knowledge and institutional barrier → Investment	0.188	0.090	0.036	Accepted
Financial barrier → Investment	−0.227	0.110	0.039	Accepted
Technical barrier → Investment	−0.150	0.072	0.036	Accepted
Energy motivation → Investment	0.146	0.084	0.083	Accepted
Financial motivation → Investment	0.351	0.078	0.000	Accepted
Environmental motivation → Investment	0.465	0.082	0.000	Accepted

Table 4.
Structural model result

These predictor variables are either continuous or categorical. Methods based on artificial intelligence such as ANNs are considered as a good predicting model in comparison to econometric models such as linear regression (Hruschka, 1993) or logistic regression model (Chiang *et al.*, 2006). Being a member of data-mining techniques, the ANN model can "allay several assumptions of statistical techniques, including the issues of linearity, multicollinearity, and normally distributed data" making the tool perform more effectively in comparison to traditional statistical tools (Garver, 2002).

The ANN model comprises three layers – input layer (independent layer), output layer (dependent layer) and hidden layer, which lay between these two layers (Figure 2). Based on the human nervous system, ANNs contain small units called neurons. The relationships among these neurons determine the network system. One such common network system is the "back propagation gradient network", which has the advantage of using potential data with "lesser processing time". This tool found extensive application in various fields of management, human resource, energy, economics and many others.

In the present context, this technique shall facilitate in deciphering the complex linear and/or non-linear relationships among the predictor variables and rooftop PV adoption. As a technique based purely on the ANN was found inappropriate for testing the causality, so the present study applied the sequential use of two complementary techniques. In the first stage, SEM was applied to confirm the relationships between six antecedent variables (FM; ENM; EVM; technical barrier; financial barrier and KNINB) and investment decision (investment in rooftop PV as dependent variable).

Data analysis

To analyse the ANN technique, the sample data set (322) was randomly segmented into two parts – training set and a test set. With 71.7% of the respondents, the elements of the training group were used to determine possible predictive associations and to develop a predictive model. The test group (28.3% of the data set) was used to validate the model.

Results and analysis

A cross-entropy function-based method was used for increasing the overall performance of the ANN model, and it also helps in reducing stagnation periods. From Table 5, we note that the overall incorrect prediction in the training set is 31.2%, which further reduced to 23.1% in the test set.

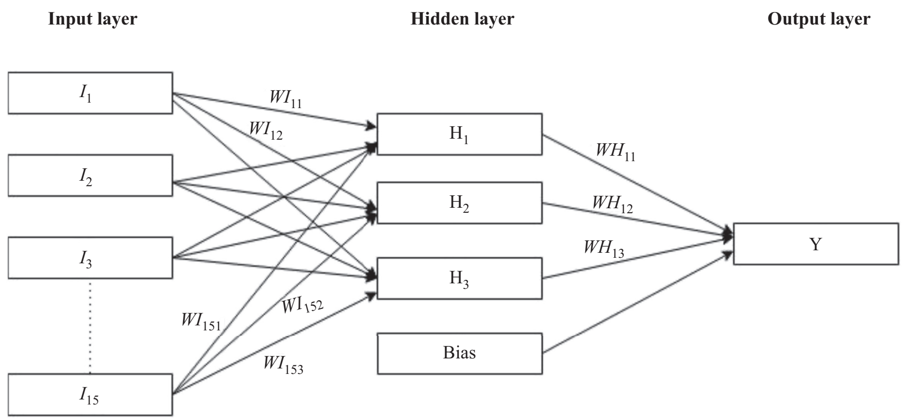

Figure 2.
Neural network

Sensitivity analysis

The prime objective of this study is to identify the most relevant factors that have a strong influence on the respondent investment decision for solar panels as a dependent variable. To achieve this objective sensitivity analysis is also conducted for identifying the important factors. Many studies suggested that in the process of sensitivity analysis, variables that have minimal impact on the dependent variables (investment in rooftop PV) shall be removed (Engelbrecht and Cloete, 1996; Prashar *et al.*, 2015). As shown in Table 6, the maximum and minimum absolute sensitivity values are 0.851 and 0.040, respectively. It specifies that none of the value of the factors is either equal to zero or near zero in all hidden layers, hence, there was no need to remove any factor(s) from the analysis (Chiang *et al.*, 2006). Finally, data from the test set was used to analyse the predictive accuracy.

Classification matrix

Prashar *et al.* (2017) posited that the worth of any predicted model depends on the accuracy in the predictions forecasted from that model. This is calculated by "the fraction of correct predictions made". The predicting accurateness of ANN model is presented through a classification matrix. The results obtained from ANN model using two randomly selected data sets (training and testing) are presented in Table 7. From the table, it is observed that 159 out of 231 respondents (based on the training set) had their investment decision identified accurately by the model; however, only 72 were wrongly classified. Hence, the overall

Table 5.
Model summery

Training set	Percent incorrect predictions	31.2%
Test set	Percent incorrect predictions	23.1

Table 6.
Sensitivity analysis

Predictor	H(1:1)	H(1:2)
FM	0.459	–0.379
EVM	–0.505	0.300
ENM	0.279	–0.040
FB	0.398	0.382
KNINB	–0.741	–0.525
TB	–0.788	–0.851

Note(s): FM: Financial Motivation, KNINB: Knowledge and Institutional Barrier, FB: Financial Barrier, TB: Technical Barrier, ENM: Energy Motivation, EVM: Environmental Motivation

Table 7.
Classification matrix for training and test set

		Predicted		
Sample	Observed	No	Yes	Percent correct
Training	No	66 (TN)	55 (FP)	54.5%
	Yes	17 (FN)	93 (TP)	84.5%
	Overall Percent	35.9%	64.1%	68.8%
Testing	no	32 (TN)	13 (FP)	71.1%
	yes	8 (FN)	38 (TP)	82.6%
	Overall Percent	44.0%	56.0%	76.9%

Note(s): Dependent variable: Investment decision, TN – true negative, FP – false positive, FN – false negative, TP – true positive accuracy of the model's prediction is 68.8%. With the data from the test group, 71 out of 91 respondents were predicted accurately concerning their investment decision. Thus, this neural network model has an accuracy of 76.9%. Therefore, approximately 77 prospective customers of solar adaptation out of every 100 were correctly identified by the model, either they invest in solar or not.

Another significant indicator is precision or positive predictive value (PPV), which is used for measuring the goodness of a model, by calculating the ratio of forecasted true prediction (TP) to total true prediction cases (see Table 8). With 74% of precision, the ANN is considered as a good precise model. Therefore, it reflects that 74 predictions will turn out to be true out of every 100 forecastings made from this model. Furthermore, sensitivity (or recall) indicates the model capability to appropriately recognize the true positive cases (those were also predicted to be invested in solar energy) out of the overall respondents considering for investing in solar energy (Powers 2011). The model's sensitivity value is 82.61%, which reflects its good capacity in recognizing respondents considering investing in solar energy. (See Table 8)

The specificity index is used for identifying the model capability in predicting true negative cases. It describes the condition, wherein the prediction model correctly recognizes the non-investor respondents for solar energy ("no") among all the negative results (non-investor). Therefore, specificity index signifies the ANN as a prediction model ability in avoiding false alarms. With 71.11% specificity value exhibits, the ANN model was able to identify more than 71% of non-investors that was truly a negative case.

Additionally, to check the goodness of a model, the F1 score is calculated. F1 value near to or equal to "1" specifies that model is a good fit and score approaching near to "0" characterized as a bad fit. The F1 score of our ANN is 0.784 (See Table 8), showing the good fit of the model. As stated earlier all four indices – accuracy, precision, sensitivity and F1 score illustrate that the ANN as a prediction model is a good fit. Hence, the forecasting made by the model is accurate.

Importance and significance of select factors

The significance of the independent variable is the extent of change in the network's model predicted value for varying values of the variable. Table 9 shows the importance of all six factors that predict respondents considering investing in solar energy panel equipment. Most significant among these is the technical barrier (40.78%) factor. With 32.56%, knowledge and awareness emerged as a second leading factor in identifying the investor investing decision. This was followed by the financial barrier, which contributed approximately 12% of the total influence in impacting the investor investing decision. Besides these leading barriers and awareness factors, financial, environment and energy motivation factors contribute only 14.68% influencing the investment in solar energy panel equipment's investing behaviour.

Indicators	Calculation	Result (%)
Accuracy	(TP + FP)/(TP + FP + FN + TN)	56.04
Positive predictive value (PPV)	TP/(TP + FP)	74.51
Negative predictive value (NPV)	TN/(TN + FN)	80.0
Recall or sensitivity (TPR)	TP/(TP + FN)	82.61
Specificity (TNR)	TN/(TN + FP)	71.11
False-positive rate (FPR)	FP / (TN + FP)	28.90
False-negative rate (FNR)	FN / (TP + FN)	17.40
*F*1 score	2 × (PPV × Recall)/(PPV + Recall)	78.40

Table 8. Calculated indicators using test set

Discussion

The present study analyses the investment decisions of a large sample of household investors, to identify the main determinants of their choices. The present study refers to the psychosocial motivators and inhibitors of the rooftop PV investment decision process to fit the broader socio-economic context in which they are deployed. Investment patterns in solar energy are of special interest, and the present study was done to analyse the behavioural pattern of household investors in the renewable energy investment decision-making process.

Based on two waves of data collection, the findings revealed that households considering the adoption of solar panels were likely influenced by six distinct factors, three motivators and three inhibitors. The three motivating factors are financial motivators, energy and environmental motivators. Some of the primary motivators of rooftop PV investment are – households are highly motivated in investing due to energy reliability and energy independence**;** households are highly motivated in the adoption of solar panels due to the initial subsidies given by the government.

Further, three inhibitors are lack of knowledge and institutional barrier, technical and financial barriers. Some of the inhibitors of rooftop PV investment are – households are facing difficulty in finding knowledgeable companies for investing in solar**;** households think that there is uncertainty in ROI**;** it's very hard to find information for households on how to interpret the benefits of excess energy sold back to the grid based on law; difficulty for the households to compare costs between different solar panels; lack of household knowledge regarding the benefits or costs of solar panels; and households facing a problem in proper maintenance of solar panels**.** These findings were also supported by our ANN results. Aforestated barriers (financial and technical), knowledge and awareness together play a significant role in influencing the investment decision in rooftop PV. As shown in Table 9, these three factors contribute nearly 85% importance in determining/influencing the investment decision. β Values of -0.227 and -0.150 for financial and technical barriers, respectively, also indicate the power of their influence in reducing the investor investing decision. However, good knowledge and awareness can be enhanced; the investor investing decision with β 0.188 also signifies its importance.

SPSS 20 is used for modelled ANN, all significant independent variables from SEM were used as the input layer (i.e. FM; EVM; ENM; financial barrier; knowledge and awareness and technical barrier), whereas investment decision is used as the output layer. Sigmoid function as suggested by Leong *et al.* (2013) is used as an activation function for neurons in both hidden and output layers.

In terms of predicting the investors in the renewable energy investment decision, the ANN model uncovered that all six significant variables from the SEM were found significant in predicting investors in the renewable energy investment decision. As stated earlier in the result and analysis part, ANN model has high prediction accuracy. ANN model demonstrated that knowledge and awareness and technical barrier played significant enormous roles in

Table 9.
Relative significance

Variables	Importance	Importance %
Financial motivation	0.049	4.89
Environmental motivation	0.068	6.76
Energy motivation	0.030	3.03
Financial barrier	0.120	11.97
Knowledge and awareness	0.326	32.56
Technical barrier	0.408	40.78

predicting the investors in the renewable energy investment decision while FM and ENM play the least importance in predicting the same.

Theoretical implications

The findings are closely aligned with the risk–return framework of Wüstenhagen and Menichetti (2012). In line with the risk–return framework, findings revealed that government subsidies act as a primary motivator which helps in overcoming the initial risk of investment in the new technology. Further, low prices and low cost of maintenance are some of the financial motivators which may likely mitigate the long-term apprehension of returns and maintenance cost. The past researches opined that dual-policy interventions in the form of spreading awareness of environmental preservation and PV subsidies complete the cycle of policy interventions (Bürer and Wustenhagen, 2008, 2009; Lüthi and Wustenhagen, 2011). The present paper findings that the strongest motivators of PV investment come from the environmental motivator and financial motivator in the form of PV subsidies further solidify the role of policy interventions in the investment decision.

The normalized Beta scores in Table 4 indicate that motivators are stronger than barriers in rooftop PV adoption by the households. These findings have important implications for the path dependence theory of bounded rationality perspective on new investment decisions. Past researchers in PV adoption found that investors are still not able to overcome the barriers and they are still in the stage where they are stuck with fossil fuel technology due to technical and financial barriers and motivators have lesser importance in PV adoption. Studies have identified evidence for path dependence in the newly emerging renewable energy sector; discuss effects of barriers on Finland's attempts to diversify its economy away from fossil fuels; barriers can also lead to lock-in; and it has been suggested that the world is currently locked into a high-carbon economy, due to barriers which are difficult to transcend; and path-dependent barriers preclude oil companies to shift from the previous path of carbon energy to renewable energy which is less treaded (Wustenhagen and Menichetti, 2012; Lovio *et al.*, 2011.

The results of our study that motivators are stronger than barriers in rooftop PV adoption by the households are also different from a recent study done in Santiago, Chile, where authors found that "strength of barriers exceeds that of motivations for households to invest in a PV technology and further observed that this aligns with what has been observed in Santiago, namely, a negligible uptake of PV energy technologies in households" (Walters *et al.* 2018, p. 1280). In the case of Kerala, India, as per government figures, the state is already advancing in the adoption of rooftop PV, as pointed by some of the recent announcements of the government of Kerala (Times of India, Kochi edition, Feb, 2019). The findings also corroborate that individual investors in Kerala are no longer stuck at the barrier stage, rather motivators are more important factors for them. Therefore, it can be concluded that as the individuals advance in rooftop PV adoption stages, the path-dependent barriers are no longer stronger than motivators. The motivators start breaking the path-dependent barriers and pave way for PV adoption.

Practical implications

In rural areas still, solar energy is not a viable option at households. So it will need out-of-box thinking. Solar vitality hardware conveys part of complex adornments which is troublesome for a typical man to get it. So it makes troublesome circumstance for him to get persuaded and receive (PV module, charge controller, battery, etc.). Because of this complexity, they do not know how it will get corrected when the system breaks down because of some fault – the upfront cost. To overcome this, EMI kind of mechanism should be brought in place at minimum interest. Solar energy devices are not popular because energy storage mechanism is very uneconomical and portability is highly compromised (lead-acid battery is used as an energy storage device). Because of this, it does not give the feeling of energy security to the users when compared to 1 gallon of gasoline gives when it is with users.

Overall, domestic PV usage in Kerala, India, presents a compelling urban case study of a theoretically ideal consumer base that has achieved some acceptance of household PV technology but still away from the full adoption of rooftop PV. Not many such examples are present in the literature. Given the relative significance of these motivators and barriers in the given context of Kerala, India, existing literature points to some targeted policy solutions which can work in this context for such a consumer base. Past literature on renewable energy policy suggests that policy formulation for such consumer base as that of Kerala should focus more on motivators such as competitive prices in the form of subsidies, providing multiple vendors with different specifications and designs and running advertisement and awareness campaigns on energy independence and environmental responsibility (Rai *et al.*, 2016; Zhang *et al.*, 2015). Still, they should do this without losing an eye on knowledge, institutional and technical barriers.

Conclusion

According to the study made, four factors have been analysed as the behavioural factors affecting investment in solar adoption in Kerala. Households are getting motivated for the adoption of solar panels as well as they are facing barriers such as financial, institutional and technical barriers. It could be beneficial to develop and advertise communication platforms or workshops to educate users and increase the visibility of successful projects. This study presents a compelling urban case study of a theoretically ideal consumer base that has not yet achieved significant uptake of household PV technology. The findings point to the household's adoption being inhabited by high upfront costs and uncertainty in financial returns as barriers. Building on this existing work highlights the complex interplay among social and institutional factors that influence a household's decision to invest in solar PV technology. This study can help shape ongoing research and practice that seeks to improve solar technology pricing and payment schemes to create a tipping point for the solar energy transition. The government should get more involved in making households understand the benefits of having solar panels. Solar panel companies must be able to guide the households clearly regarding the number of solar panels to be installed.

Limitations

The study used two data sets of 237 households and 387 households making the results fairly robust and generalizable. However, the method may have some limitations also which the future studies could eliminate. Data from primary sources may have response bias, primarily due to social desirability, which has been taken care of in the present study. Though the present study has a large sample size, it was in no way large enough to cover all segments of the customer. Therefore, the sample size is a limitation in this study. Future, the study could use clustering and cover enough customers from all segments.

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